Restaurant Segmentation using Reviews and Metadata

Abstract

This project focuses on applying machine learning techniques to segment restaurants based on customer reviews and business metadata. The rapid growth of online food platforms has resulted in massive volumes of both structured and unstructured data. While these data sources contain valuable insights about customer preferences, pricing strategies, and operational efficiency, they are often underutilized due to their complexity and scale.

The primary objective of this project is to develop a data-driven framework that transforms raw restaurant data into meaningful clusters. By combining natural language processing on customer reviews with numerical and categorical restaurant attributes, the project uncovers hidden patterns that can support strategic decision-making.

The methodology includes data preprocessing, feature engineering, outlier handling, dimensionality reduction, and the implementation of multiple clustering algorithms such as K-Means, DBSCAN, and Hierarchical Clustering. Each model is evaluated using internal validation metrics including Silhouette Score, Davies–Bouldin Index, and Inertia. Based on these evaluations, the most effective clustering model is selected and analyzed for business relevance.

This report provides a complete end-to-end pipeline for restaurant segmentation and demonstrates how machine learning can be used to enhance marketing strategies, pricing optimization, and customer experience in the hospitality industry.

1. Introduction

The restaurant and hospitality industry is highly competitive and data-intensive. With the increasing adoption of online food platforms, restaurants now generate large amounts of data in the form of customer reviews, ratings, follower counts, pricing information, and operational details. However, much of this data remains underutilized due to the lack of effective analytical tools that can convert raw data into actionable business insights.

Traditional approaches rely on manual analysis or basic reporting, which often fail to capture complex relationships between customer perception and business performance. For example, understanding how pricing affects ratings, how service hours influence popularity, or how different cuisines attract different customer segments requires advanced analytical techniques.

Machine learning, particularly unsupervised learning, offers a powerful approach to discover natural groupings in data without requiring pre-labeled outcomes. Clustering techniques can group restaurants with similar characteristics, enabling stakeholders to identify market segments, design targeted promotions, optimize operations, and enhance customer experience.

This project leverages both structured metadata and unstructured textual reviews to build a comprehensive restaurant segmentation model. By integrating natural language processing with numerical feature analysis, the study provides a holistic view of restaurant performance.

2. Problem Statement

Restaurants and food aggregators face significant challenges in understanding diverse customer preferences and competitive positioning. Despite the availability of extensive review data and operational metrics, businesses struggle to translate this information into meaningful segments that can guide strategic decisions.

The core problem addressed in this project is how to transform heterogeneous restaurant data into well-defined clusters that reflect similar business characteristics and customer perceptions. Specifically, the project seeks to answer the following questions:

- How can unstructured customer reviews and structured business attributes be cleaned, standardized, and integrated into a single analytical dataset?
- Which clustering algorithm provides the most effective segmentation in terms of cluster separation, compactness, and interpretability?
- How can the resulting clusters be interpreted to deliver tangible business value in marketing, pricing, and operational planning?

Addressing these challenges will enable data-driven restaurant segmentation that supports

targeted marketing campaigns, improved customer satisfaction, and enhanced operational efficiency.

3. Dataset Description

Two datasets were utilized in this project: a restaurant metadata dataset and a customer reviews dataset. Together, these datasets provide both quantitative and qualitative insights into restaurant operations and customer experiences.

The restaurant metadata dataset includes attributes such as restaurant name, average cost, cuisine types, operating hours, total number of reviews, number of followers, and the number of pictures uploaded. These features describe the business profile of each restaurant.

The customer reviews dataset contains individual reviews, ratings, reviewer identifiers, timestamps, and engagement metrics such as pictures per review. This dataset provides direct insight into customer satisfaction and sentiment.

To ensure meaningful clustering, review-level data was aggregated to the restaurant level, preventing duplication bias from restaurants with a large number of reviews. This resulted in a unified dataset where each row represents a single restaurant with consolidated performance indicators.

Name	Links	Cost	Collections	Cuisines	Timings
Beyond Flavours	https://www.zomato.com	800	Food Hygiene Rated Restaurants i	Chinese, Continental, Kebab, Eur	12noon to 3:30pm, 6:30pm to 11:30pm (Mon-Sun)
Paradise	https://www.zomato.com	800	Hyderabad's Hottest	Biryani, North Indian, Chinese	11 AM to 11 PM
Flechazo	https://www.zomato.com	1,300	Great Buffets, Hyderabad's Hottes	Asian, Mediterranean, North Ind	11:30 AM to 4:30 PM, 6:30 PM to 11 PM
Shah Ghouse Hotel & Restaurant	https://www.zomato.com	800	Late Night Restaurants	Biryani, North Indian, Chinese, S	12 Noon to 2 AM
Over The Moon Brew Company	https://www.zomato.com	1,200	Best Bars & Pubs, Food Hygiene Ra	Asian, Continental, North Indian	12noon to 11pm (Mon, Tue, Wed, Thu, Sun), 12noon to 12midnight (Fri-Sat)
The Fisherman's Wharf	https://www.zomato.com	1,500	Sunday Brunches, Corporate Favo	Seafood, Goan, North Indian, Co	12Noon to 3:30PM, 4PM to 6:30PM, 7PM to 11:30PM (Mon, Tue, Wed, Thu, Sun), 12
eat.fit	https://www.zomato.com	500	Food Hygiene Rated Restaurants i	Healthy Food, North Indian, Con	7 AM to 10 PM
Shah Ghouse Spl Shawarma	https://www.zomato.com	300		Lebanese	12 Noon to 12 Midnight
Hyper Local	https://www.zomato.com	1,000	Great Buffets, Live Sports Screenir	American, Chinese, North Indian	12 Noon to 12 Midnight
Cream Stone	https://www.zomato.com	350	Frozen Delights	Ice Cream, Desserts	10 AM to 1 AM (Mon-Thu), 10 AM to 1:30 AM (Fri-Sun)
Sardarji's Chaats & More	https://www.zomato.com	400	Street Food, Veggie Friendly, Pock	Street Food, North Indian, Fast F	12 Noon to 3:30 PM, 7 PM to 10:30 PM
Barbeque Nation	https://www.zomato.com	1,600	Barbecue & Grill	Mediterranean, North Indian, Ke	12 Noon to 3:30 PM, 6:30 PM to 11:30 PM
Absolute Sizzlers	https://www.zomato.com	750	Great Buffets	Continental, American, Chinese	11:30 AM to 1 AM
The Lal Street - Bar Exchange	https://www.zomato.com	1,300	Corporate Favorites, Happy Hours	North Indian, Italian, Finger Foo	12noon to 12midnight (Mon-Sun)

Restaurant	Reviewer	Review	Rating	Metadata	Time	Pictures
Beyond Flavours	Rusha Chakraborty	The ambience was good, food was quite good . had	5	1 Review , 2 Followers	5/25/2019 15:54	0
Beyond Flavours	Anusha Tirumalaneedi	Ambience is too good for a pleasant evening. Service is ver	5	3 Reviews , 2 Followers	5/25/2019 14:20	0
Beyond Flavours	Ashok Shekhawat	A must try.. great food great ambience. Thnx for the servic	5	2 Reviews , 3 Followers	5/24/2019 22:54	0
Beyond Flavours	Swapnil Sarkar	Soumen das and Arun was a great guy. Only because of the	5	1 Review , 1 Follower	5/24/2019 22:11	0
Beyond Flavours	Dileep	Food is good.we ordered Kodi drumsticks and basket mutto	5	3 Reviews , 2 Followers	5/24/2019 21:37	0
Beyond Flavours	Nagabhavani K	Ambiance is good, service is good, food is aPradeecp and	5	1 Review	5/24/2019 15:22	0
Beyond Flavours	Jamuna Bhuwalka	Its a very nice place, ambience is different, all the food we	5	1 Review	5/24/2019 1:02	0
Beyond Flavours	Sandhya S	Well after reading so many reviews finally visited this place	4	1 Review	5/23/2019 15:01	0
Beyond Flavours	Akash Thorat	Excellent food , specially if you like spicy food . Courteous	5	1 Review , 1 Follower	5/22/2019 23:12	0
Beyond Flavours	Smarak Patnaik	Came for the birthday treat of a close friend. Perfect place	5	1 Review , 1 Follower	5/22/2019 22:37	0
Beyond Flavours	Saubhagya Bhuyan	The service was great and the food was awesome. The ser	5	1 Review , 1 Follower	5/22/2019 22:22	0
Beyond Flavours	Srivaths07	Very good ambience, amazing food ,good service and frien	5	3 Reviews , 1 Follower	5/22/2019 21:43	0
Beyond Flavours	Kunj Mishra	Food was very good. Soup was as expected. In starters we	5	1 Review , 1 Follower	5/22/2019 21:29	0
Beyond Flavours	Pradeep Vetapalem	Food is too good. Telangana kodiak fry is must try.Mutton	5	2 Reviews , 25 Followers	5/22/2019 21:21	0
Beyond Flavours	Kankaria.ritu	We ordered corn cheese balls, manchow soup and paneer	1	1 Review	5/22/2019 21:19	0
Beyond Flavours	Abhay Sharma	Food and ambience is fantastic.. Waiter Manav Maji was re	5	4 Reviews , 7 Followers	5/22/2019 20:38	0
Beyond Flavours	Shubham Jaiswal	Came here for lunch and the food was good and tasty.	4	30 Reviews , 34 Followers	5/22/2019 19:54	8
Beyond Flavours	Srijani Mukherjee	The best thing about this place is the food. My favorite go t	5	6 Reviews , 267 Followers	5/22/2019 19:47	1
Beyond Flavours	Hari Jangam	Polite and friendly staff. Nice ambience and good spread o	5	1 Review , 1 Follower	5/22/2019 15:13	0
Beyond Flavours	Suneet Soni	Food is really good. We had vegetarian items including nan	5	1 Review , 33 Followers	5/21/2019 22:08	0

4. Data Preprocessing

Data preprocessing is a critical step to ensure data quality and model reliability. The raw datasets contained missing values, inconsistent formats, and redundant information that required systematic cleaning.

Missing numerical values, such as cost and rating, were imputed using median values to minimize the impact of extreme observations. Categorical variables representing cuisine groups and operational features were encoded using binary encoding, where the presence of a feature was represented by 1 and its absence by 0. Textual fields with missing values were removed, as meaningful imputation for free-text data is not feasible.

```
hotel_review = hotel_review.dropna(subset=['Review', 'Rating'])
```

```
hotel_review['Total Followers'] = hotel_review['Total Followers'].fillna(0)
hotel_review['Total Review'] = hotel_review['Total Review'].fillna(0)
```

```
num_cols = ['Cost', 'Rating', 'open_duration_hours']

for col in num_cols:
    restaurant_df[col] = restaurant_df[col].fillna(restaurant_df[col].median())
```

```
binary_cols = [
    'late_night','breakfast_service','opens_early',
    'closes_late','split_shift'
]

restaurant_df[binary_cols] = restaurant_df[binary_cols].fillna(0)
```

```
cuisine_cols = [
    'is_Indian','is_Asian','is_Middle_Eastern','is_Western',
    'is_Fast_Food','is_Desserts','is_Healthy','is_BBQ_Grill'
]

restaurant_df[cuisine_cols] = restaurant_df[cuisine_cols].fillna(0)
```

Duplicate records were identified and removed to ensure that each restaurant was represented uniquely in the dataset. These steps established a consistent and reliable dataset for subsequent analysis.

```
df2.drop_duplicates(inplace=True)
print(f"Duplicate Rows in df2 : {df2.duplicated().sum()}")
```

5. Textual Data Preprocessing

Customer reviews are unstructured and require extensive natural language processing (NLP) before they can be used for analysis. Several preprocessing steps were applied to standardize and clean the textual data.

The text was first converted to lowercase to ensure uniformity. Punctuation marks, URLs, and numerical digits were removed to reduce noise. Stopwords such as 'the', 'and', and 'is' were eliminated to retain only meaningful terms. Tokenization was then performed to break reviews into individual words.

Lemmatization was applied to convert words into their base forms, preserving semantic meaning while reducing dimensionality. Finally, TF-IDF vectorization was used to transform text into numerical representations. TF-IDF assigns higher importance to words that are frequent in a document but rare across the corpus, making it ideal for capturing meaningful patterns in customer feedback.

```
from nltk.tokenize import word_tokenize

hotel_review['tokens'] = hotel_review['Review'].apply(word_tokenize)
```

```
lemmatizer = WordNetLemmatizer()

def lemmatize_text(tokens):
    return [lemmatizer.lemmatize(word) for word in tokens]

hotel_review['lemmatized_tokens'] = hotel_review['tokens'].apply(lemmatize_text)
```

```
# Vectorizing Text
hotel_review['clean_text'] = hotel_review['lemmatized_tokens'].apply(lambda x: " ".join(x))
```

```
tfidf = TfidfVectorizer(max_features=1000)

tfidf_matrix = tfidf.fit_transform(hotel_review['clean_text'])
```

6. Feature Engineering

Feature engineering was performed to enhance the representational power of the dataset. Raw attributes were transformed into higher-level features that better capture restaurant characteristics.

Cuisine information was grouped into broader categories such as Indian, Asian, Western, Fast Food, Desserts, Healthy, and BBQ. Binary indicators were created for each cuisine category to allow multi-label representation.

```
cuisine_groups = {
    'Indian': Indian,
    'Asian': Asian,
    'Middle_Eastern': Middle_Eastern,
    'Western': Western,
    'Fast_Food': Fast_Food,
    'Desserts': Desserts,
    'Healthy': Healthy,
    'BBQ_Grill': BBQ_Grill
}

def map_cuisine_group(cuisines):
    if pd.isna(cuisines):
        return 'Unknown'
    cuisines = [c.strip() for c in cuisines.split(',')]
    groups = set()
    for g, clist in cuisine_groups.items():
        if any(c in clist for c in cuisines):
            groups.add(g)
    return ', '.join(groups) if groups else 'Others'

df1['Cuisine_Group'] = df1['Cuisines'].apply(map_cuisine_group)
```

Operational timings were processed to derive features such as opening hour, closing hour, total operating duration, breakfast service, late-night service, and split shifts. These features capture service availability and business model patterns.

```python
def parse_timings(timing):
    if pd.isna(timing):
        return None, None, 0

    timing = timing.lower()
    timing = re.sub(r'\(.*?\)', '', timing)

    slots = timing.split(',')
    opens = []
    closes = []

    for s in slots:
        parts = s.split('to')
        if len(parts) != 2:
            continue

        o = time_to_hour(parts[0])
        c = time_to_hour(parts[1])

        if o is not None and c is not None:
            opens.append(o)
            closes.append(c)

    if not opens or not closes:
        return None, None, 0

    return min(opens), max(closes), len(slots)

df1[['open_hour', 'close_hour', 'num_slots']] = (
    df1['Timings']
    .apply(lambda x: pd.Series(parse_timings(x)))
)
```

Additional composite metrics were created, including Popularity Score (based on total reviews and followers) and Value for Money (rating divided by cost). These engineered features provide more meaningful insights into customer perception and business performance.

7. Outlier Handling

Outliers can significantly distort clustering results by disproportionately influencing distance calculations. Therefore, systematic outlier detection and treatment were performed.

The Interquartile Range (IQR) method was used to identify extreme values in numerical features such as cost, number of reviews, and number of followers. Instead of removing

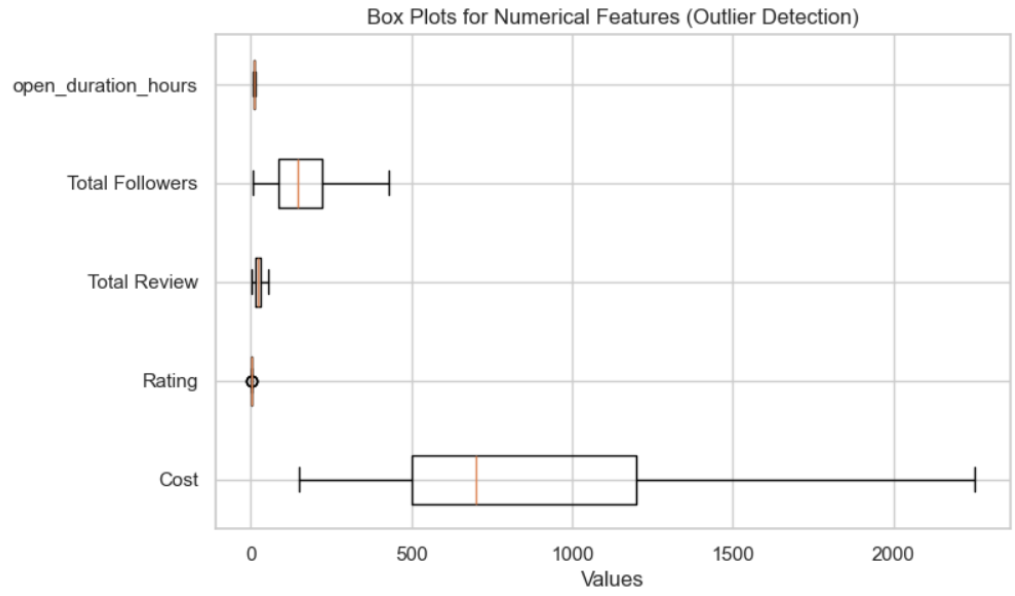
these records, which could result in data loss, values were capped within acceptable bounds to preserve legitimate but extreme observations.

```
def cap_outliers_iqr(df, col):
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    df[col] = df[col].clip(lower, upper)

for col in ['Cost','open_duration_hours','Total Followers','Total Review']:
    cap_outliers_iqr(restaurant_df, col)
```

```
restaurant_df['Cost'] = restaurant_df['Cost'].clip(
    restaurant_df['Cost'].quantile(0.01),
    restaurant_df['Cost'].quantile(0.99)
)

restaurant_df['Total Followers'] = restaurant_df['Total Followers'].clip(
    restaurant_df['Total Followers'].quantile(0.01),
    restaurant_df['Total Followers'].quantile(0.99)
)
```

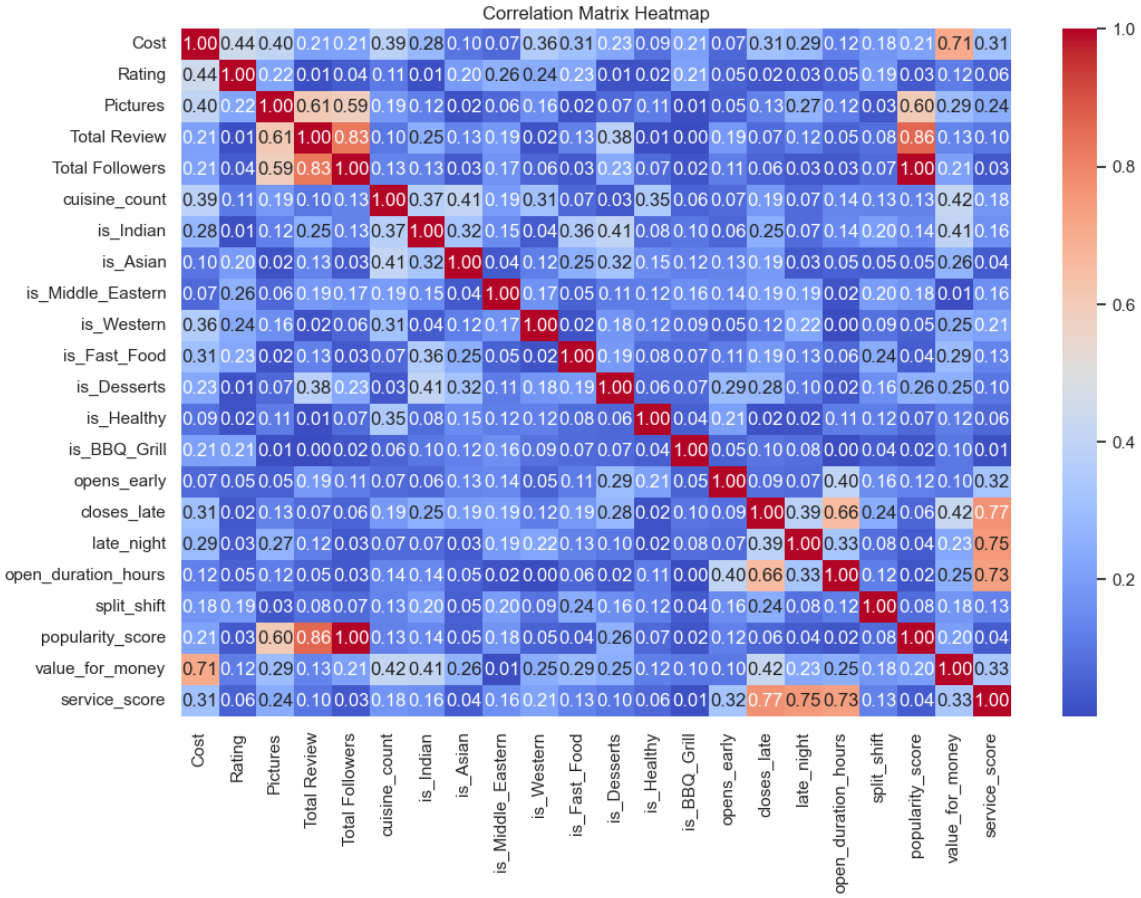


8. Feature Selection

To reduce redundancy and improve clustering efficiency, feature selection techniques were applied. Highly correlated variables were identified using correlation analysis and removed to prevent multicollinearity.

Low-variance features that contributed little to differentiation were eliminated. Domain knowledge was also incorporated to retain only business-relevant attributes such as pricing, customer satisfaction, popularity metrics, cuisine types, and operational features.

This process resulted in a compact, interpretable feature set suitable for unsupervised learning.



9. Data Scaling and Transformation

Clustering algorithms based on distance metrics are sensitive to feature magnitude. To ensure that all features contribute equally, numerical variables were standardized using Z-score normalization through StandardScaler.

This transformation ensures that features with large numeric ranges do not dominate distance calculations, leading to more balanced and reliable clustering outcomes.

10. Dimensionality Reduction (PCA)

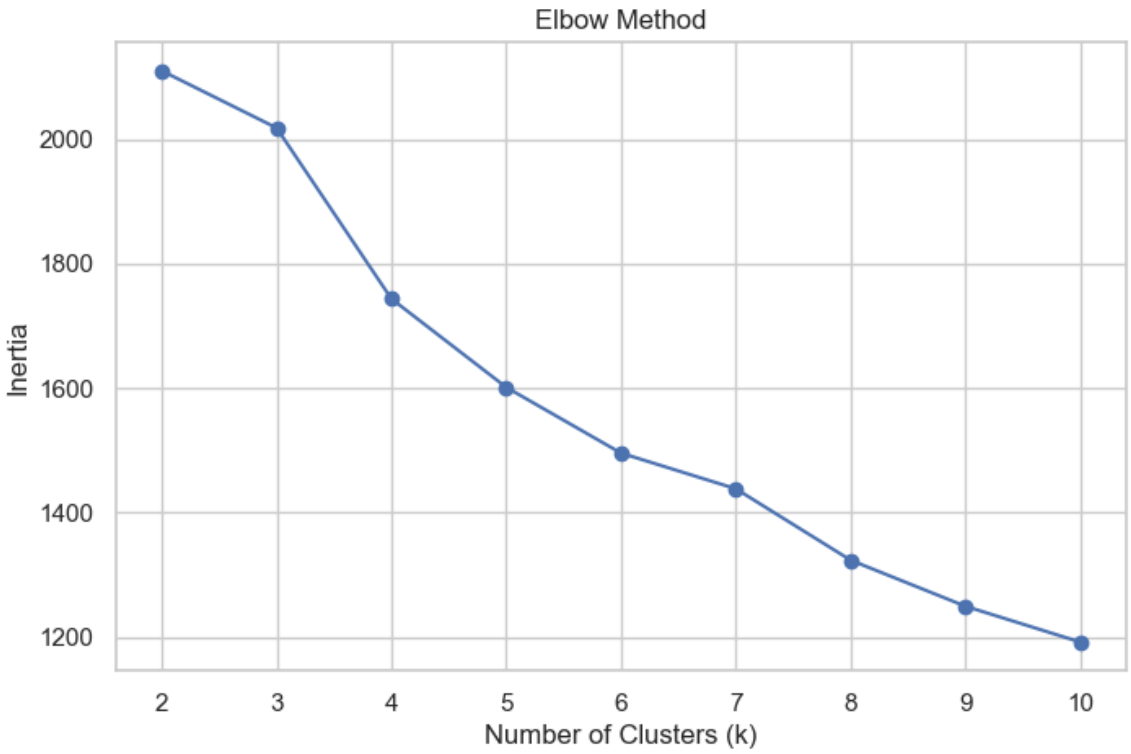
Principal Component Analysis (PCA) was applied to reduce dimensionality for visualization purposes. PCA transforms the original features into a set of orthogonal components that capture the maximum variance in the data.

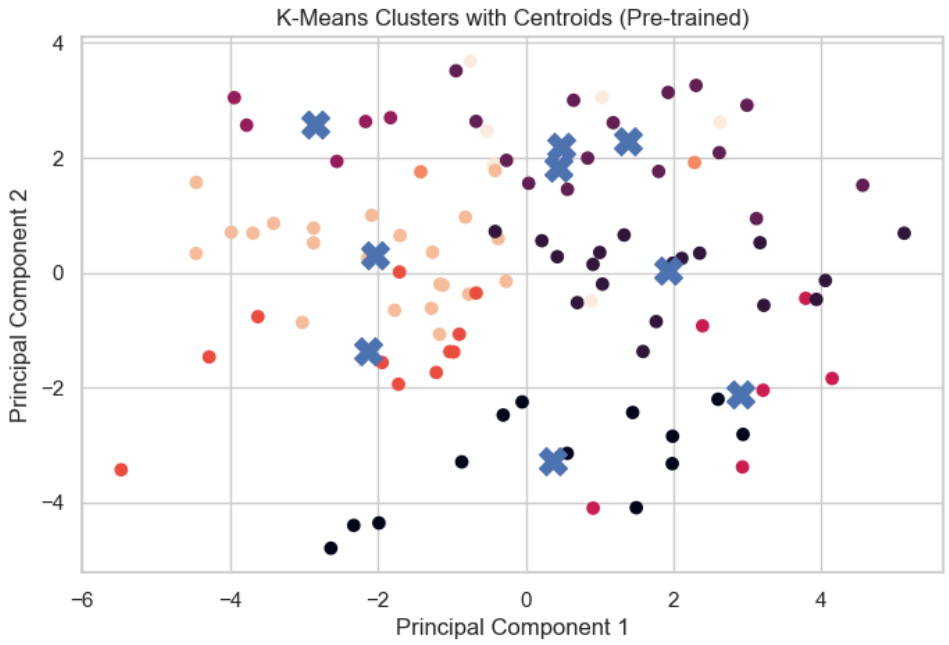
By projecting the high-dimensional dataset into two principal components, cluster structures could be visualized in a two-dimensional space without retraining the model. This facilitated intuitive interpretation of restaurant groupings.

11. Model 1: K-Means Clustering

K-Means clustering was implemented as the primary segmentation model. The algorithm partitions the dataset into a predefined number of clusters by minimizing within-cluster variance.

Model performance was evaluated using Silhouette Score, Davies–Bouldin Index, and Inertia. Hyperparameter tuning was conducted to determine the optimal number of clusters, resulting in improved segmentation quality. K-Means demonstrated strong cluster separation and compactness, making it the most effective model for this dataset.

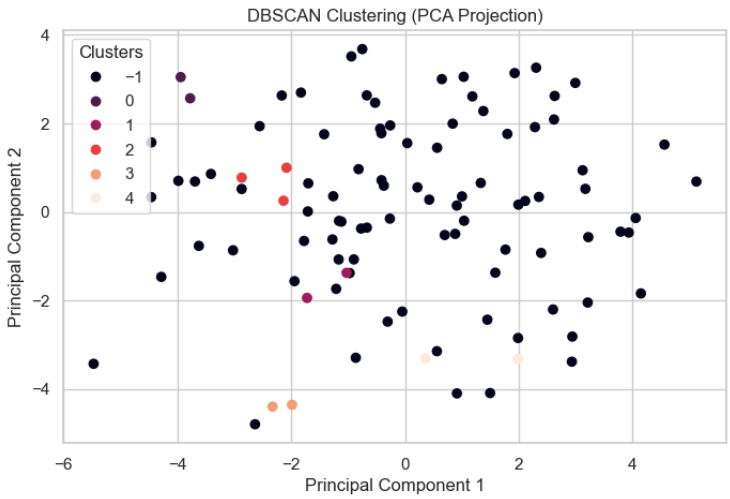




12. Model 2: DBSCAN

DBSCAN, a density-based clustering algorithm, was tested as an alternative approach. Unlike K-Means, DBSCAN identifies clusters of arbitrary shape and detects noise points automatically.

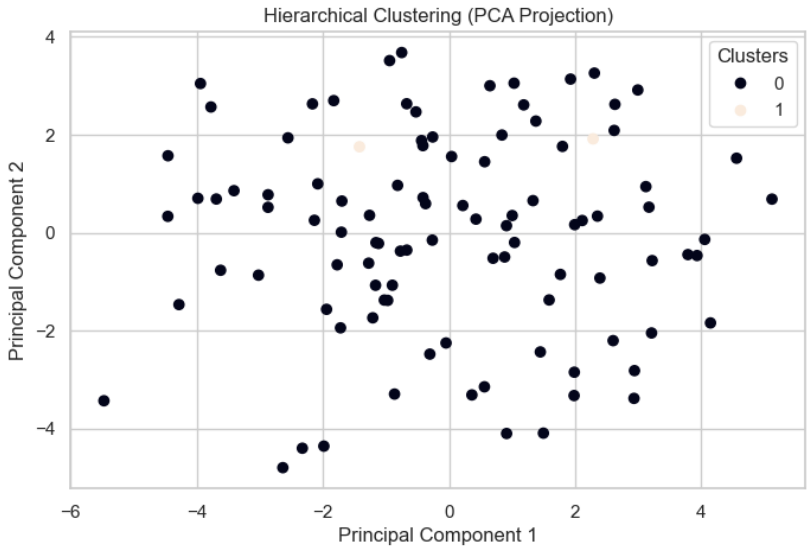
However, in this project, DBSCAN classified most data points as noise, indicating that the dataset does not exhibit strong density-based structures. This outcome highlights the importance of selecting algorithms based on data characteristics and reinforces the suitability of K-Means for this problem.



13. Model 3: Hierarchical Clustering

Hierarchical clustering was implemented using the agglomerative approach, which builds clusters in a bottom-up manner. This technique provides a tree-like representation of data relationships.

Although hierarchical clustering produced interpretable clusters, it formed fewer and less distinct segments compared to K-Means. Evaluation metrics indicated weaker separation, making it less suitable as the final model.



14. Model Comparison

All three models were compared using Silhouette Score and Davies–Bouldin Index. K-Means consistently achieved the highest separation and lowest compactness error. DBSCAN failed to identify meaningful clusters, while hierarchical clustering produced overly broad groups. Based on quantitative metrics and qualitative interpretability, K-Means was selected as the final model.

	Model	Silhouette Score	Davies-Bouldin Index	Clusters Formed
0	K-Means	0.130096	1.724113	9
1	DBSCAN	NaN	NaN	5
2	Hierarchical	0.119662	1.854393	2

15. Model Explainability

Since clustering is unsupervised, model explainability was achieved through centroid-based analysis. By examining the variance of feature values across cluster centroids, the most influential variables were identified.

Key drivers of segmentation included cost, customer ratings, popularity metrics, cuisine categories, and service availability. This explainability ensures transparency and builds trust in the model's outputs.

16. Business Impact

The final clustering solution offers substantial business value. Restaurants can be grouped into segments such as budget-friendly, premium, highly popular, niche, or service-oriented establishments.

These segments enable targeted marketing, personalized recommendations, pricing strategy optimization, operational improvements, and identification of underperforming restaurants. The framework supports data-driven decision-making in the hospitality sector.

17. Conclusion

This project presented a comprehensive pipeline for restaurant segmentation using machine learning. By integrating textual review analysis with structured business attributes, the study identified meaningful restaurant segments.

Among the evaluated models, K-Means emerged as the most effective due to its superior performance, scalability, and interpretability. The proposed framework demonstrates how data science can transform raw customer feedback and business data into actionable insights.